Selected beam-and-block brief

BB02 / VB01 / 9 November 2026 / Supplier enquiry - NOT FOR CONSTRUCTION

The owner has selected **beam and block with a designed reinforced topping** over the basement, for **one fully laden van and one Tesla parked together**. The drawn 150 mm beams are provisional geometry only. Please design and price suitable beam sizes, spacing, topping and supports; assess 225 mm beams or another suitable arrangement. Solid RC is retained for comparison.

Owner-agreed vehicle allowance	Maximum gross mass	Static weight	Assumed heavy axle*	Assumed heavier wheel*
Laden van	3.5 tonnes	34.34 kN	26.78 kN	16.07 kN
Tesla allowance	3.1 tonnes	30.41 kN	23.72 kN	14.23 kN

Together: **6.6 tonnes / 64.75 kN static**. Both vehicles may be present. This total is not a point load or a substitute for wheel patterns. Vehicles are variable loads and are not counted as permanent resistance to groundwater uplift.

*Study assumptions: 65% of gross weight on the heavier axle; 60:40 left-right wheel split; an additional x1.20 movement sensitivity. These are not manufacturer axle ratings or prescribed code factors. Check both directions, both axles, turning and edge positions.

Separate preliminary load cases	Basis / limits
5.0 kPa vehicle UDL	Provisional Category G screening basis; not added to the vehicle point cases.
20 / 25 / 30 kN local point sensitivities	Upper 30 kN case represents a provisional floor jack action. Contact patch and jacking arrangement unconfirmed.
Paired van / Tesla wheel loads	1.6 / 1.8 / 2.0 m assumed tracks. One to five beams sharing hypothetically; no sharing assumed verified.

Design basis to confirm: both vehicle allowances exceed 30 kN. JRC first-generation guidance places this in Category G; the applicable UK National Annex, local concentrated actions, combinations and future access restrictions must be confirmed. The original low vehicle sensitivities have been superseded. Neither deck is currently a verified structural design.

Floor plan and bearing routes

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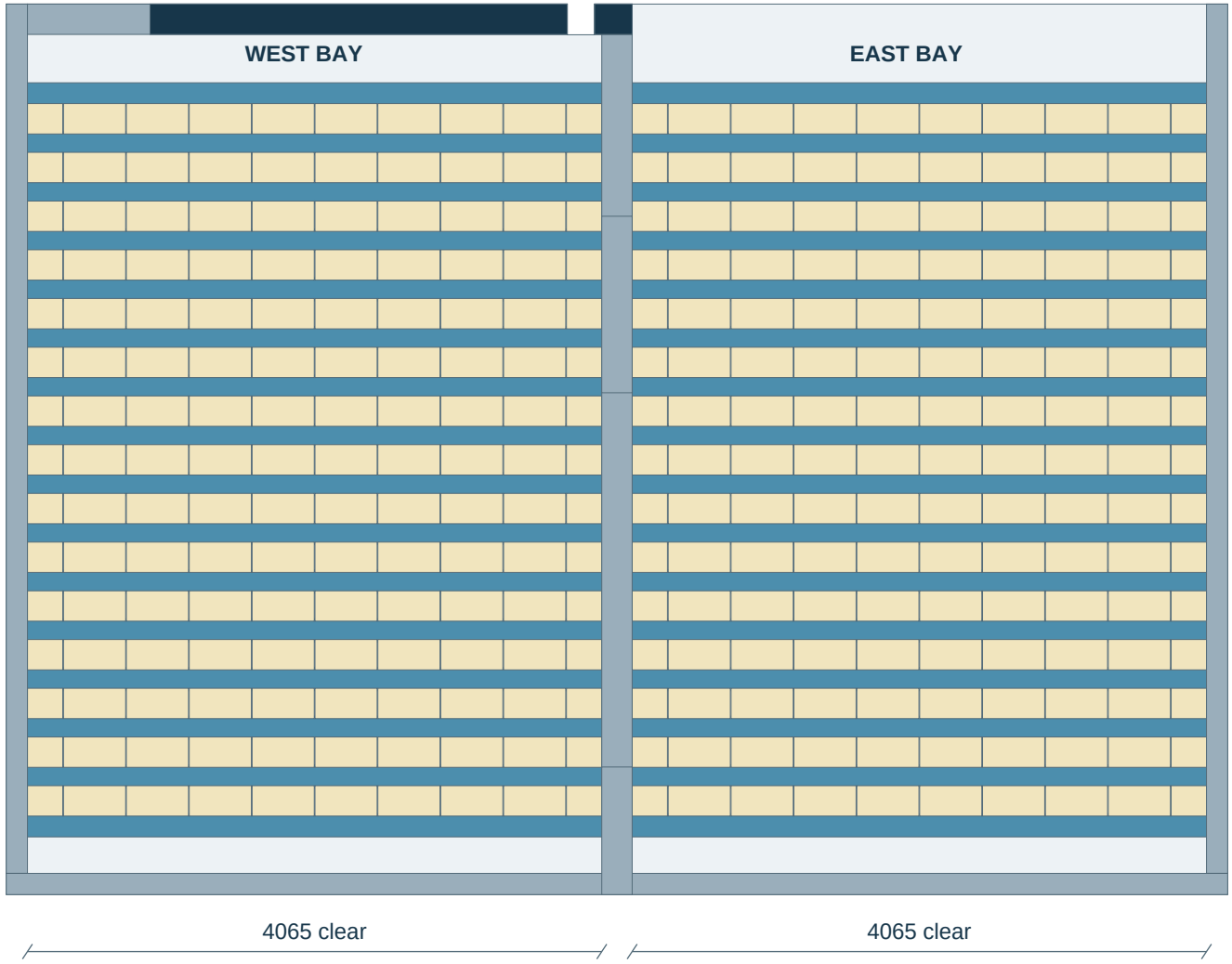
Upper bearing-wall strip / stair-girder junction



Plan / topping hidden

- 32 precast beams
- 300 infill pieces
- Concrete support below
- Upper masonry above solid strip

Beams run east-west across two short bays. Each beam is nominally 4290 mm long, at 345 mm centres.

Perimeter bearing: 125 mm.
Centre bearing: 100 mm each side of a 215 mm wall. Opposing ends have a 15 mm modelled gap.

Central support includes two doorway heads. Their reactions, bending, bearing and reinforcement require design.

The complete garage has 7 m clear depth. This drawing covers the current suspended RC deck footprint only; verify its junction with adjacent floor/threshold construction.

Block cuts, simplified beam profiles and bearing lengths are coordination geometry only. No supplier production layout or site measurement is implied.

Floor depth and bearing junctions

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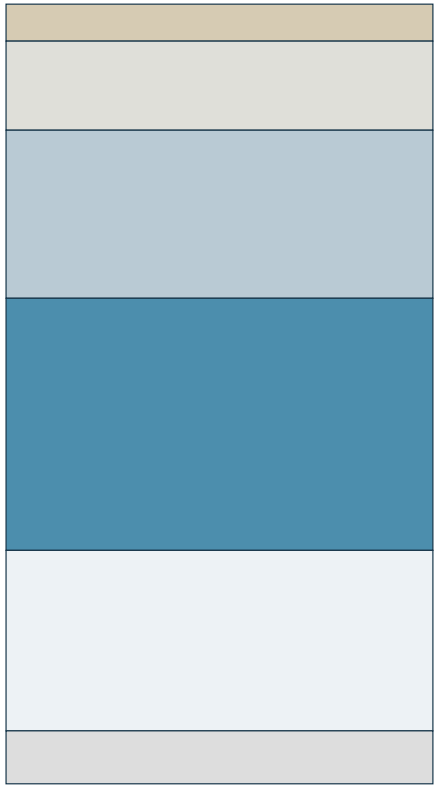
A / Provisional 150 mm geometry - levels retained

FFL 0.000 m



22 mm Finish allowance

53 mm Screed allowance

100 mm Nominal reinforced topping

150 mm Beam + infill zone

107.4 mm Service space

31.5 mm Ceiling build-up

Structural underside -0.325 m. Ceiling board top -0.4324 m; finished ceiling -0.4639 m. Nominal topping includes a 25 mm camber/tolerance budget above the 75 mm comparison allowance.

Final fire, insulation, acoustic and waterproofing layers remain to be specified and fitted. The diagram preserves current allowances; it does not certify their performance.

B / Central wall bearing

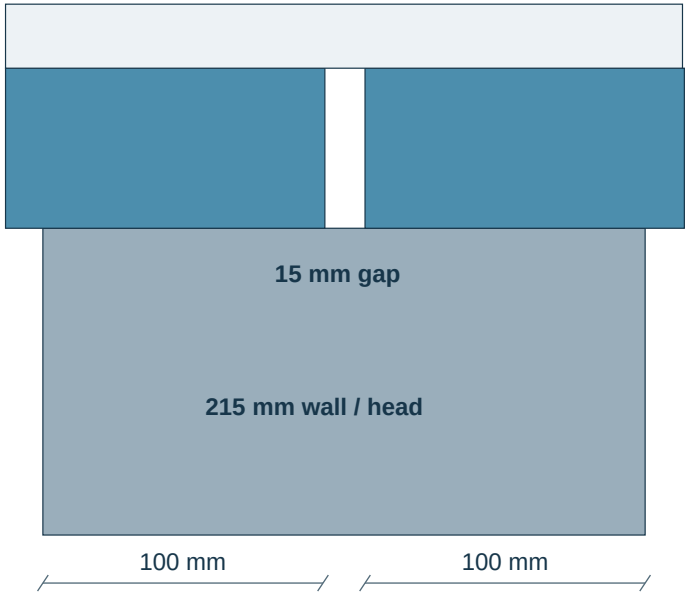


Diagram enlarged; not to scale. Grout/closure, restraint ties and wall-head reinforcement require design. The 125 mm perimeter seat needs a coordinated concrete pocket, waterproofing and edge restraint detail.

225 mm alternative: 225 beam + 100 topping + 22 finish = 347 mm. Omitting the 53 mm screed allowance lowers the bearing/soffit by 22 mm and leaves about 85 mm above the current ceiling board. Recheck all bearing and opening-head levels; not drawn or adopted.

Comparison and requested supplier return

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Quantity / interface	Solid RC alternative	Selected beam-and-block
Structural envelope	250 mm	150 mm beam + 100 mm topping
Site concrete	13.62 m3	6.96 m3 + 0.253 m3 grout
Additional precast / blocks	None	32 beams / 137.28 m; 300 infill pieces
Deck self-weight	340.5 kN	297.2 kN
Status	Unverified RC spatial candidate	Selected system; sizes unverified

Reduced site concrete is not the same as a complete installed saving. Please return separate supply-only and installed prices, delivery/lifting, reinforcement, grout/topping, local RC strips, formwork/propping, membranes and design inclusions. Identify exclusions and programme assumptions.

Required technical return: adopted UK design basis; vehicle/axle/jack load schedule; beam profile, spacing and camber; infill grade and cut layout; topping reinforcement and proven transverse load distribution; reactions at each bearing and doorway head; wall/edge ties and basement diaphragm action; construction-stage checks; fire rating and coordinated floor/ceiling build-up. Mark any change in floor height or support geometry.

Project structural engineer: coordinate the supplier reactions with concrete strips, basement walls, opening heads, raft, groundwater uplift and temporary works. A lighter deck slightly reduces ground pressure but also reduces uplift resistance. Retained upper masonry must stay on a designed support strip, not the infill.

Do not infer approval from the comparison: old supplier tables are legacy BS 8110 benchmarks. The current 150 mm option has unresolved local load-sharing demands. No bars, mesh, tendons, concrete pockets or production beam sizes are selected by this enquiry.

Sources / issue record

JRC Eurocode 1 workshop / p32 - categories and indicative actions

IStructE - UK Eurocode transition / confirm edition at design appointment

Forterra / Bison - site-specific beam-and-block design and quotation

Stressline guide / legacy reference only